

Motor Imagery Classification Using Single Channel of EEG in Online Brain-Computer Interface

Maryam Taghizadeh

Computer Engineering and Information Technology
Razi University
Kermanshah, Iran
Taghizadeh.maryam@razi.ac.ir

Abdollah Chalechale*

Computer Engineering and Information Technology
Razi University
Kermanshah, Iran
Chalechale@razi.ac.ir

Abstract—This paper presents an efficient method with simplified calculations for classifying left and right hand movement imagery using a single channel of EEG to design a brain-computer interface (BCI) system. The proposed method utilizes wavelet transform to decompose EEG signals into multiple frequency bands. The classification task is performed using the support vector machine (SVM) with the radial basis function (RBF) kernel and the K-nearest neighbor (KNN) algorithm. This paper introduces and compares two techniques that are based on the sixth and eighth levels produced from a deconstructed single C3 channel. For feature extraction, statistical information such as variance, standard deviation, and signal power are individually considered. The achieved results indicate high accuracy for left and right-hand movement, with 100% accuracy for left-hand movement and 87.47% accuracy for right-hand movement. These results were accomplished using SVM with the RBF kernel and KNN algorithms, based on power features extracted from the eighth-level EEG signal. Compared to prior methods utilizing single-channel and multi-channel approaches, this method demonstrates superior performance.

Keywords—Wavelet Transform; BCI; Classification; Motor Imagery

I. INTRODUCTION

The brain-computer interface (BCI) is a well-known way for straight real-time interact brain signals to the environment. This communication system does not adhere to the natural pathways of the brain such as muscles and nerves. Instead, it reveals the subject's commands when the subject is quietly silent and without any movement. Examining brain signals during different mental activities can be used to manage various processes. The primary purpose of the BCI system is to support patients with severe motor impairment affecting means of communication. Moreover, the BCI system has the potential to be utilized in various domains such as robotics, biomedical technology, and the development of artificial limbs and mechanical vehicles. EEG signals are widely used in non-invasive BCI devices. Thus, it can be regarded as one of the most significant signals for assessing brain functions. The EEG technique involves the process of capturing brain activity using an amplifier and filtering system to eliminate any unwanted noise. EEG signals are ordinarily recorded using Ag-AgCl electrodes, and the signal values are reported in the range of 0.5-100 μ V. Several linear and nonlinear analyses have been proposed for examining the EEG signal in order to assess sleeping stages, diagnose the level of anesthesia, recognize Alzheimer's disease [1], explore sports informatics [2], and detect epilepsy [3].

Motor imagery (MI) refers to the neural activity in the primary sensorimotor areas that resembles the execution of an actual action, even when no physical movement has occurred [4]. The BCI system based on motor imagery EEG data has attracted much attentions. Some studies specifically focused on the imagination of left-hand, right-hand, or foot movement. This study concentrates on the BCI system based on motor imagery containing hand movement. Motor imagery studies employ either multi-channel or single-channel methods. Although multi-channel EEG signals are appealing methods, they also introduce a higher level of difficulty to experiments [5]. Hence, it is indispensable to utilize BCI technology using a reduced number of channels. On the other hand, one of the significant challenges in designing online BCI is to reduce execution timing. Thus, it can be practical for real-time BCI systems to support disabled patients. To achieve this aim, our focus is on minimizing the number of channels utilized for data collection and their extracted features.

These systems usually employ power spectral, parametric models of EEG signal, and so on as feature vectors. In the following, linear discriminate analysis (LDA), neural networks, and other classifiers are utilized for feature classification. In general, four steps are required to design a BCI system: EEG data acquisition, preprocessing and feature extraction, classification, and use of the classified signal to monitor the system. Also, other features such as standard deviation, signal entropy, and co-occurrence matrix-based textural descriptors have been used. To classify, kNN, support vector machine, neural network, deep learning, and genetic algorithm have been applied. In these solutions, multiple channels are used to collect data [6, 7]. In contrast, in some proposed methods, data are acquired using a single channel. In 2014, Ge. et al proposed a hand movement classification method based on single-channel [8]. To feature extraction, short-time Fourier transform (STFT) is applied. In addition to STFT, a time-frequency texture descriptor is extracted and a logistic regression classifier is considered for left and right movement [9]. In this paper, the main aim is to utilize single-channel from EEG data to classify hand movements. Using a single channel can reduce the complexity of computation costs and improve execution time. Consequently, it is capable of producing reliable outcomes in a real-time BCI system. One of the main ideas in this paper is to reduce the number of EEG channels while maintaining an acceptable level of classification accuracy. This idea can be effective in reducing computational processes. For the analysis of left and right hand movement, we select EEG data from the C3 channel and apply two levels of decomposed wavelet transform (DWT). For the sake of clarity, we have referred to them as the proposed

method-(1) and the proposed method-(2), respectively. The multilevel wavelet transform is used to separate the frequency sub-bands and power signal, and a feature extraction algorithm applies statistical information. SVM and KNN are employed for classification, and the two proposed methods are assessed and compared to previous studies. The main contributions of this paper are:

(1). Proposing an approach based on wavelet transform in two variations, referred to as method-1 and method-2, with the objective of decreasing complexity and computation time by segregating frequency sub-bands and power signal. Moreover, statistical information is applied to extract features.

(2). The efficacy of extracted features is evaluated by investigating two classifiers, namely SVM and KNN. The method is assessed using the motor imagery dataset and compared to other methods that have been examined on the same dataset.

The rest of the paper is organized as follows. Section II provides an overview of previous studies on the classification of motor imagery using EEG signals. The two proposed methods are elaborated in detail in Section III. Experimental results are reported and analyzed in Section IV. Finally, the conclusion and future work are outlined.

II. RELATED WORKS

Movement recognition is an essential stage to design and develop BCI systems. Afterwards EEG data acquisition, feature extraction algorithm is performed. For example, mean, variance, standard deviation, short-time Fourier transform (STFT), wavelet transform, and recursive energy performance can be used. Note that two statistical techniques such as principle component analysis (PCA) and independent component analysis (ICA) can be adopted to reduce feature vector dimension. Furthermore, one of the most important issues in online BCI system design is the choice of classification method. It has to quickly classify movement imagination with more accuracy so that it can be applied in the online BCI system.

According to several reported studies, three different algorithms can be generally considered as feature extraction: power spectral analysis which shows the variance of the power signal. Adaptive automated regression (AAR) describes the variance of EEG signal corresponding to the event without previous information from related bands. This way is effective for online analysis. Another effective approach is a common spatial pattern (CSP) designing a spatial filter. The filter variances are useful to separate two classes in EEG signals [10]. Afterward, it usually uses SVM, KNN, neural network, and deep learning for classifying. For data acquisition from EEG signals, single-channel or multi-channel can be utilized. The primary aim of using multiple channels is to improve the accuracy of classification. By applying single channel as EEG input, the method reduces computation costs.

Several approaches have been suggested based on multi-channel in the BCI system. In 2011 intelligent algorithms were used to categorize left and right hand signals [6]. It utilized C3 and C4 channels from EEG. Feature vectors such as wavelet coefficients, power spectral density of the alpha, the central beta band, and the average power of the

respective bands were considered classifying. They evaluated several classification methods such as SVM with radial basis function, Bayesian, linear discriminant analysis, Quadratic discriminant analysis (QDA), k-nearest neighbor (KNN). The SVM with RBF kernel achieved the highest performance, with an accuracy of approximately 82.14%. To reduce feature dimension, the principle component analysis (PCA) method is also applied. Aowlad et al. used [11] the beta frequency band and extracted features including energy, scale variance, RMS value, Roll Off, variance, approximation entropy, zero crossing, and modified MAV. Probabilities neural network (PNN) is used to classify left/right-hand movement. The obtained outcomes by PNN indicates more accuracy in comparison to back propagation neural network. An approach based on P3 and P4 channels is introduced to classify movement imagination in [10]. The approach relies on DWT to decompose the average power of the C3/C4 and P3/P4 channels in motor imagery. The reconstructed signal from A6 on the sixth level is selected to create a feature vector. Fisher linear discriminant analysis is then executed to calculate two different thresholds. Furthermore, the performance of FLDA is compared to SVM. A different method has been developed to classify left and right-hand movements [12]. A preprocessing technique is accomplished on C3 and C4 channels. The feature vector consists of wavelet transform coefficients, welch power spectral density estimations, and the average power. PCA is employed to reduce feature vector dimensions. Next, an unsupervised technique called self-organizing maps (SOM) based on neural networks is introduced to classification. Sridhar and Manian developed a deep learning network to evaluate cognitive functions and identify relationships with brain signal features based on auditory, olfactory, movement, and motor imagery [13]. Three principal components are applied to the network for the training process. A deep convolutional neural network (DCNN) can also be used to extract spatial and frequency features and classify motor imagery tasks. The proposed DCNN is followed by long-short term memory (LSTM) [14]. Unlike the previous studies, the study presented in [15] concentrated on false positive reduction relying on time-frequency features in the BCI system. To achieve the next generation of motor imagery brain-computer interface in autonomous wearable systems, a new model based on EEGNet [16] is provided in [17] to reduce memory footprint with benefit accuracy.

Some studies have reported on the motor imagery classification using a single channel of EEG input. Here, we provide a concise overview of the relevant studies conducted on the single channel of EEG. For a more comprehensive understanding, readers can refer to the study [5]. In [8], the suggested method is based on time-frequency analysis and using short-time Fourier transform. Concerning this idea, the obtained signal is transformed to the time-frequency domain and it creates some informative channels. After reconstruction data, the common spatial pattern (CSP) feature is used and the SVM classifier is employed to train data. The highest accuracy is obtained for C4 approximately 88.1%. To compute spectrogram from EEG signal, Short time Fourier transform (STFT) is used [9]. The co-occurrence matrix is next estimated of gray level values from the image input. The texture features such as correlation energy, contrast, homogeneity, and dissimilarity are extracted. To train, logistic regression is applied. This idea is assessed through offline and online conditions, and it

achieves an accuracy of 87.6% for the offline case. Chen et al. presented two models to feature extraction in the single channel motor imagery. The models given are derived from an improved version of a two-dimensional nonlinear FitzHugh-Nagumo neuron system [18]. Table I. summarizes the related works with their specific details.

This study aims to enhance the accuracy of classifying motor imagery by employing a single channel from EEG data, thus reducing the need for complicated computations. First, frequency bands are split using multilevel wavelet transform [19]. Second, the separated coefficients are used to extract features using the proposed method-(1) and proposed method-(2). Subsequently, two different classification algorithms, SVM with RBF kernel and KNN are applied.

III. PROPOSED METHOD

This section first describes EEG data. Then, the proposed method with details is elaborated.

A. EEG Data

The EEG signal is able to reveal both the normal and abnormal electrical activity of the brain. Consequently, it has been employed as a very powerful tool in the field of neurology and clinical neurophysiology. This signal consists of different frequency bands that can be effective in identifying mechanisms. Brain waves are categorized into four basic groups: delta (0-4 Hz), theta (4-8 Hz), alpha (8-13 Hz), and beta (13-30 Hz) [20]. The amplitude is measured within the range of 0.5 to 100 μ V. Theta frequency is around 20 μ V, while the alpha frequency ranges from 30 to 50 μ V. Beta waves usually indicate low amplitude in the range of 5 to 30 μ V. These waves carry information from different brain activities. A frequency band between 1 to 35 Hz is a preferable range of motor imagery classification. The used electrodes can have either 14 or 64 channels. The EEG signal is collected using electrodes placed on the scalp which consists of many waves with different characteristics and can represent special and vital information. To simulate and imagine the motor function in the BCI system, it usually relies on C3, C4, and Cz channels [21, 22]. In this paper, the proposed method is implemented using given data from C3.

B. Proposed Algorithm Steps

Due to the fluctuating nature of EEG throughout time, it has evolved into a non-stationary signal. Because of this characteristic, time-frequency approaches are employed. Wavelet transform is a very popular technique for noise removal and analysis. Wavelet decomposition effectively captures and precisely localizes transient features, in both the time and frequency domains. A large time window represents low-frequency information, while a small time window indicates high-frequency information. Note that more variations are observed at high frequencies. Firstly, in the proposed method, multilevel wavelet transform is employed to divide the EEG data into several band frequencies. Daubechies wavelet by eight levels is applied as shown in Fig. 1. Then, we consider D6 and A8, which correspond to the sixth and eighth levels, for feature extraction.

To feature extraction, three components are considered: the power average signal, standard deviation, and variance. The power average feature is considered a preferable feature

TABLE I. SUMMARY OF REVIEWED PAPERS

Method	Feature	Classifier	Channel	Benefit
[6]	Wavelet, power spectral density of alpha, central beta band, average power of respective bands	SVM, KNN, QDA, LDA, Bayesian	C3, C4	Simple, robust
[11]	Energy, scale variance, RMS, variance, entropy	PNN	Multi	Fast training, Speed up classification
[10]	DWT	FLDA	C3/C4, P3/P4	High classification rate
[12]	Wavelet, Welench power spectral density, average power	SOM	C3/C4	Robust, efficient
[14]	DCNN	DCNN, LSTM	Multi	Increase accuracy
[17]	EEGNet	CNN model	Multi	Memory footprint reduction
[8]	Time-frequency, STFT, CSP	SVM	Single	Address four classes by single channel
[9]	STFT, texture, GLCM	Logistic regression	Single	Efficient
[18]	Non-linear FitzHugh-Nagumo	SVM, KNN, RF, LDA	Single	Robustness
Our method	Multilevel wavelet transform	SVM, KNN	Single	Reduce Complexity and time

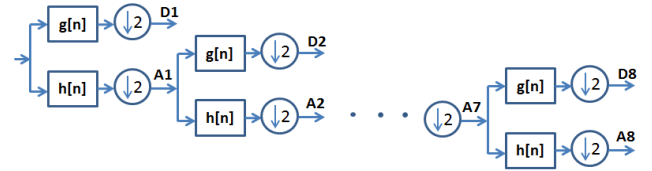


Fig. 1. Decomposed signal by wavelet transform

for EEG signal classification. To compute power signal, the energy of the signal is first calculated and it is next divided into time ranges. The power functions for continuous and discrete signal are (1) and (2).

$$E = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} |x(t)|^2 dt \quad (1)$$

$$E = \frac{1}{(n_2 - n_1 + 1)} \sum_{n=n_1}^{n_2} |x[n]|^2 \quad (2)$$

The variance parameter is defined as the average of the squared differences from the mean. It can be expressed as (3):

$$\delta^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \quad (3)$$

Standard deviation (SD) is equal to the square of the variance. It shows the distance between data and average value on average, and it is computed by (4):

$$S = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2} \quad (4)$$

Afterwards, estimated feature vectors are created, and we use two classification methods: SVM and KNN. For two different classes for left and right movement, using the SVM classifier can be effective and it indicates high generalization. The SVM algorithm directly explores the separating hyperplane with largest margin. In this method, data can be separated into linear and non-linear. For the linearly separable case, it can be denoted as (5):

$$g(x) = \text{sign}(w, \theta(x) + b) \quad (5)$$

Clearly, it determines positive and negative classes. The vector w is perpendicular, b is the distance between the closest points and separating line and x is the feature vector. For non-linear data, SVM utilizes non-linear kernels and maps data into space with more dimensions making them linearly separable in that space. The most common kernels in SVM include polynomial, radial basis function (RBF), and sigmoid. In this case, the decision function based on the kernel is established as (6):

$$g(x) = \text{sign}(\sum_{i=1}^N \alpha_i y_i k(x_i, x) + b) \quad (6)$$

Where, $i=\{1,2,\dots,N\}$, $x_i \in \mathbb{R}^d$, and d is a feature vector dimension, N denotes the number of training data according to two classes, $y_i \in \{-1, +1\}$ and α_i is a coefficient that is computed using optimization problems. Because of non-linear characteristics of the EEG signal we employ the radial basis function kernel.

Another classifier is the k-nearest neighbor (KNN). Under this framework, all instances are considered as points in an n -dimensional space, and proximity between them is determined by the Euclidian distance. The k parameter describes the number of neighbors. In this study, k is equal to one. In this case, the distance between x_i and x_j is given by (7):

$$d(x_i, x_j) = \sqrt{\sum_{r=1}^n (a_r(x_i) - a_r(x_j))^2} \quad (7)$$

IV. EXPERIMENTAL RESULTS

To verify the proposed method, we measure two important parameters: classification rate and execution time which are acquired using an Intel Core i7 system. The proposed method is evaluated by the motor imagery dataset that is provided by the NUST laboratory [23]. The subject is 21 years old with a right-handed male without known diseases. The acquired data contains the movement of the left and right hand recorded with eyes closed randomly. In total, different 19 channels are used and AC lines work at 50 Hz and the sampling rate is 500 Hz. As mentioned above, we have selected electrode C3 for our assessments.

To assess the proposed method, each three components of the feature vector based on two different levels is considered as training by two classifiers. Also, we train each feature distinctly. The best outcomes resulted in the power signal. To detect left-hand movement based on the sixth level, the SVM classifier is shown the best reliable result. The obtained results for the sixth level with each of three

feature vectors and two classifiers are illustrated in Table II. We assumed a one-second long signal for each trained and test data and the number of test data is 32800 samples. Note that the best execution time also corresponds to the power signal. To detect right-hand movement, the SVM classifier is also employed. The summary of acquired results is listed in Table III. The number of test data is 31800 samples. The average variance parameter for left and right-hand movement is 96.39%. Due to the longer execution time compared to the power feature, we finally selected findings produced using signal power.

In the proposed method based on the eight level, we consider 32300 samples to test, and the length of each signal is equal to 3200 samples. The outcomes are summarized in Table IV for both left and right-hand movements. The SVM and KNN classifiers acquired the highest classification rate for left and right movement respectively. Note that the KNN classifier imposes more computations in comparison with SVM.

Fig. 2 compares the proposed method and the study [9]. Both of the proposed methods demonstrate high performance, with the size of their feature vectors being lower than [9]. The length of the feature vector dimension for proposed method-(1), proposed-method-(2), and the previous work [9] are 65, 20, and 120, respectively.

Moreover, Fig. 3 compares various classifiers for multi-channel methods and the proposed method.

It illustrates reasonable classification rate for the single channel in SVM and SVM+KNN in compared with previous methods based on multi-channel.

V. CONCLUSION

This paper examined the classification of left and right-hand movements based on a single channel. The proposed method is based on extracting two different frequency bands

TABLE II. RESULTS OF SIXTH LEVEL WITH SVM FOR LEFT HAND

	Power	Standard deviation	Variance
Classification rate	99.72	67.07	99.70
Execution time (μ s)	385	397	397

TABLE III. RESULTS OF SIXTH LEVEL WITH SVM FOR RIGHT HAND

	Power	Standard deviation	Variance
Classification rate	90.57	41.51	93.08
Execution time (μ s)	380	408	377

TABLE IV. RESULTS IN EIGHTH LEVEL FOR LEFT AND RIGHT MOVEMENT

	classification	Execution time (μ s)
Left hand (with SVM)	100	375
Right hand (with KNN)	87.5	654

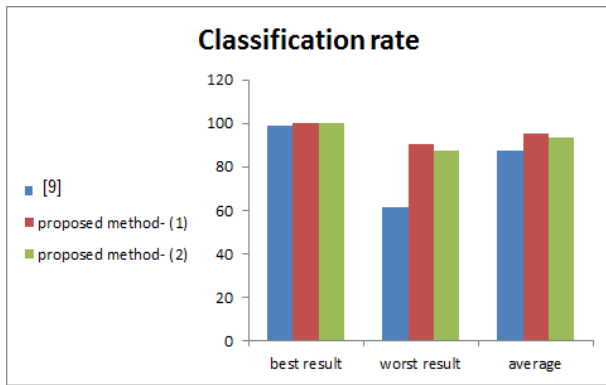


Fig. 2. A compaision between proposed method-(1), (2), and [9]

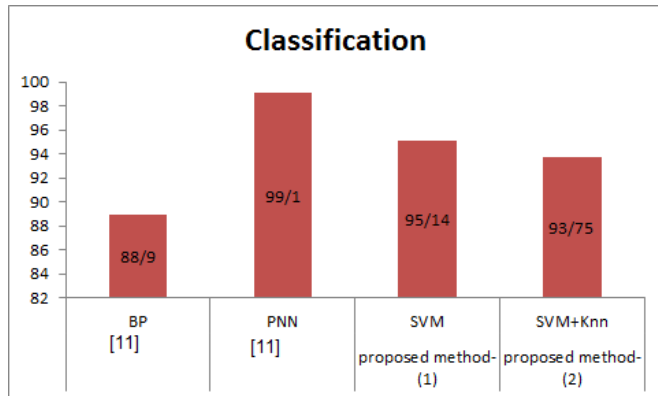


Fig. 3. Clasification rate for multi-channel and the proposed method

associated with multilevel wavelet transform. For feature extraction, statistical information, including variance, standard deviation, and power average, is assumed and analyzed. The data is trained using two classification algorithms: KNN and SVM with a RBF kernel. The findings obtained from the SVM classifier showed that the highest accuracy was achieved for left-hand motions at the sixth and eighth levels. Additionally, the SVM and KNN classifiers confirmed good accuracy for right-hand movements at the sixth and eighth levels, respectively. Furthermore, the optimal result is directly related to the signal's power. Moreover, it demonstrated a decrease in the amount of time required for execution. In compared to previous methods whether multi-channel or single channel, our proposed method represents high accuracy. This proposed method can be considered suitable for designing an online BCI system due to its low complexity. In the future, to enhance accuracy, packet wavelet transforms and a combination of the features can be considered. We will evaluate the proposed method in the online BCI system empirically.

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